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Enhancing Supply Chain Efficiency through Predictive Analytics: A Systematic Approach to Forecast Unit of Activity

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Summary

The supply chain of the oil and gas industry is known for its complexity and uncertainty. Lasker (2022) conducted a comprehensive study to examine the challenges of the industry's supply chain and the potential of data analytics to improve its performance. Our own supply chain faces similar difficulties, particularly in predicting Unit of Activity (UoA) such as transaction volume, which currently relies on inconsistent and manual methods due to decentralized data and diverse measurement approaches.

Existing approaches do not provide a systematic way of predicting UoA, and prone to inaccuracies and inefficiencies. To address these issues, this study proposed predictive analytics with a combination of statistical methods and machine learning models to forecast purchase order transaction volume.

According to Lee et al. (2022), predictive analytics utilizes data, statistical techniques, and machine learning algorithms to predict future events and trends, ultimately improving organizational performance. The proposed methodology follows a defined process, including project definition, data collection, data analysis, statistical analysis, modelling, deployment, and model monitoring.

By embracing predictive analytics, proposed approach offers a more efficient and reliable method for UoA prediction, contributing to optimal supply chain performance.

Introduction

The supply chain within the oil and gas industry is renowned for its intricate nature and the multitude of operational activities it entails, all of which are accompanied by significant levels of uncertainty. Recognizing these complexities, Laskar (2022) undertook a comprehensive study to delve deeper into the challenges inherent in the industry's supply chain. The study's primary objective was to explore the potential of data analytics in addressing these challenges and enhancing overall supply chain performance. In line with the industry's complexities, our own supply chain faces similar hurdles. Specifically, the current process of predicting Unit of Activity (UoA) namely transaction volume relies on inconsistent and manual approaches. This inconsistency stems from the decentralized nature of data, which is derived from multiple tools and systems. Moreover, various subject matter experts employ diverse methods for UoA measurement and estimation, further exacerbating the challenge.

The existing approaches utilized in UoA prediction fail to provide a systematic framework that ensures accuracy, scalability, and sustainability in the long run. Consequently, inefficiencies and inaccuracies persist within the system, hindering optimal supply chain performance. Recognizing these shortcomings, it is crucial to embrace an improved methodology that leverages advanced techniques and tools to streamline UoA prediction and enhance the overall supply chain process.

To address the aforementioned issues, this study proposed employing predictive analytics with a combination of statistical approaches and machine learning models to predict purchase order transaction volumes, providing a systematic, auditable, transferable, scalable, and sustainable approach to UoA forecasting. According to Lee et al. (2022), predictive analytics leverages input data, statistical techniques, and machine learning algorithms to forecast the likelihood of specific events occurring and predict future trends or outcomes. Its ultimate goal is to enhance the performance of organizations by deriving insights into future trends or outcomes from existing data. The procedures involve define project, data collection, data analysis, statistical analysis, modelling, deployment and model monitoring (Lee et al., 2022).

Unit of Activity (UoA) Analytics

Unit of Activity (UoA) prediction is considered a regression problem because it involves estimating quantifiable measures which is continuous numerical values such as transaction volume (Seyedan & Mafakheri, 2020). Regression models are well-suited for UoA prediction as they can capture the complexity and variability of the data. In this study, the proposed approach utilizes future activities, such as rig count, type of rig, and activity rate, as drivers to predict UoA trends. This helps identify and plan around critical elements that impact efficiency and cost in supply chain operations.

To ensure the temporal consistency of the data and mimic real-world scenarios accurately, the data is split into training and test sets by year instead of using a random ratio. Three years of historical data (2019-2021) are used to train the model, while a test set including more recent data (2022) is used for evaluating model performance. This approach allows the assessment of how well the model generalizes to future data.

Before evaluating the model performance, Spearman correlation analysis is conducted to assess the correlation score between rig count and transaction volume. The analysis result showed high correlation score between rig count and transaction volume. This indicates a strong association between the variables, highlighting the reliability of rig count as an indicator for predicting transaction volume in the context of UoA prediction.

The R2 score is utilized as an evaluation criterion to assess model performance. A higher R2 score suggests a larger proportion of variance in the UoA that can be explained by the predictor (rig count), indicating a stronger ability to accurately predict UoA values. The tested models in this study include

XGBoost, Ordinary Least Squares (OLS), Lasso (L1) regression, Ridge (L2) regression, Isolation Forest, K Nearest Neighbors (KNN), and Random Forest. Among all the models, Random Forest demonstrates a high R2 score (see Figure 1). It outperforms other models by effectively capturing nonlinear relationships between input variables and the target variable. Random Forest is also robust to outliers and noisy data, making it reliable for UoA prediction. Additionally, it offers feature importance analysis and performs well with high-dimensional data, addressing challenges faced by other models.

Regression Models	**R2 Score**
XGBoost	0.818
Ordinary Least Squares (OLS)	0.418
Lasso (L1) Regression	0.409
Ridge (L2) Regression	0.417
Isolation Forest	0.577
K Nearest Neighbors (KNN)	0.779
Random Forest	0.834

Figure 1 *R2 Score Comparison of Various Regression Models*

The prediction results of transaction volume using Random Forest model achieve an impressive accuracy rate of 89%. Accuracy is calculated by dividing the number of correctly predicted instances by the total number of instances. These predictions provide valuable insights into overall trends and fluctuations in activity levels, enabling informed decision-making and effective resource allocation across various hubs based on the estimated transaction volumes. This strategic approach to resource allocation and planning minimizes bottlenecks, maximizes efficiency, and enhances the overall performance of the supply chain.

Framework Overview

All in all, the organizational strategy is generated at highest level, incorporating inputs from various participants and supply chain activities through multiple support functions (see Figure 2). To enable predictive analytics, data from the supply chain environment is utilized. The process of predicting transaction volume through predictive analytics with machine learning begins by defining the Unit of Activity (UoA) and collecting data from multiple sources while adhering to proper business rules. Data analysis is then conducted to assess the quality of the data sources. Statistical analyses, including correlation analysis (Spearman), are performed to evaluate the relationships among different factors. After extensive evaluation of feature correlations and model performance, the random forest model is selected and implemented due to its ability to handle complex data and generate robust predictions. The prediction outputs are visualized using Power BI, which aids in result interpretation and supports strategic decision-making within the organization. The insights obtained through predictive analytics are valuable for performance measurement and continuous improvement, ultimately enhancing the overall efficiency of the supply chain.

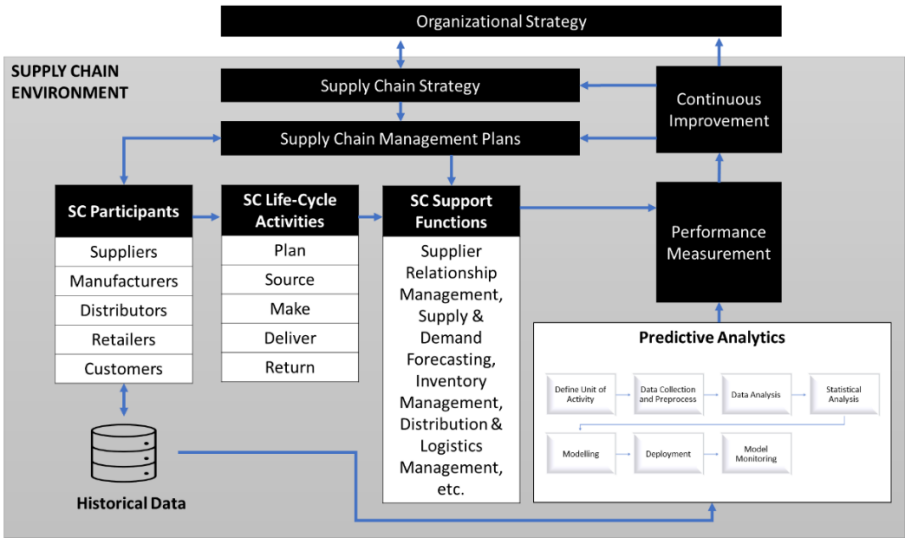

Figure 2 *Proposed Supply Chain's Predictive Analytics Framework (Adapted from Ghazal & Alzoubi, 2022; Hye et al., 2019; Lee et al., 2022)*

Conclusions

In conclusion, proposed systematic approach of application of predictive analytics on supply chain unit of activity can provide significant benefits to the supply chain by improving the accuracy and efficiency of supply chain operational activities. Accurate transaction volume prediction enables better resource planning, budgeting, and capacity management, leading to enhanced operational performance and strategic decision-making capabilities. The identified studies (Brandtner, 2022; Ghazal & Alzoubi, 2022; Wang et al., 2016; Lee & Kwon, 2018; Seyedan & Mafakheri, 2020) highlight the relevance and applicability of predictive analytics in different aspects of supply chain management, further supporting the feasibility and value of the proposed approach. This study covered transaction volume prediction as its scope. Future research can be extended to more UoAs such as payment audit, shipment processing, vendor invoice management, etc.

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